BOOST

Boost Topology:

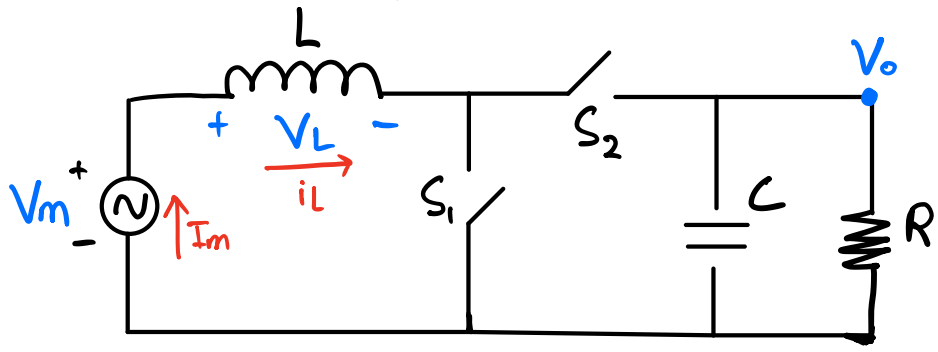
$\Delta i_L = 2A$

$V_m = 5V$

$V_o = 12V$

$P_o = 11V$

$f_s = 200kHz$



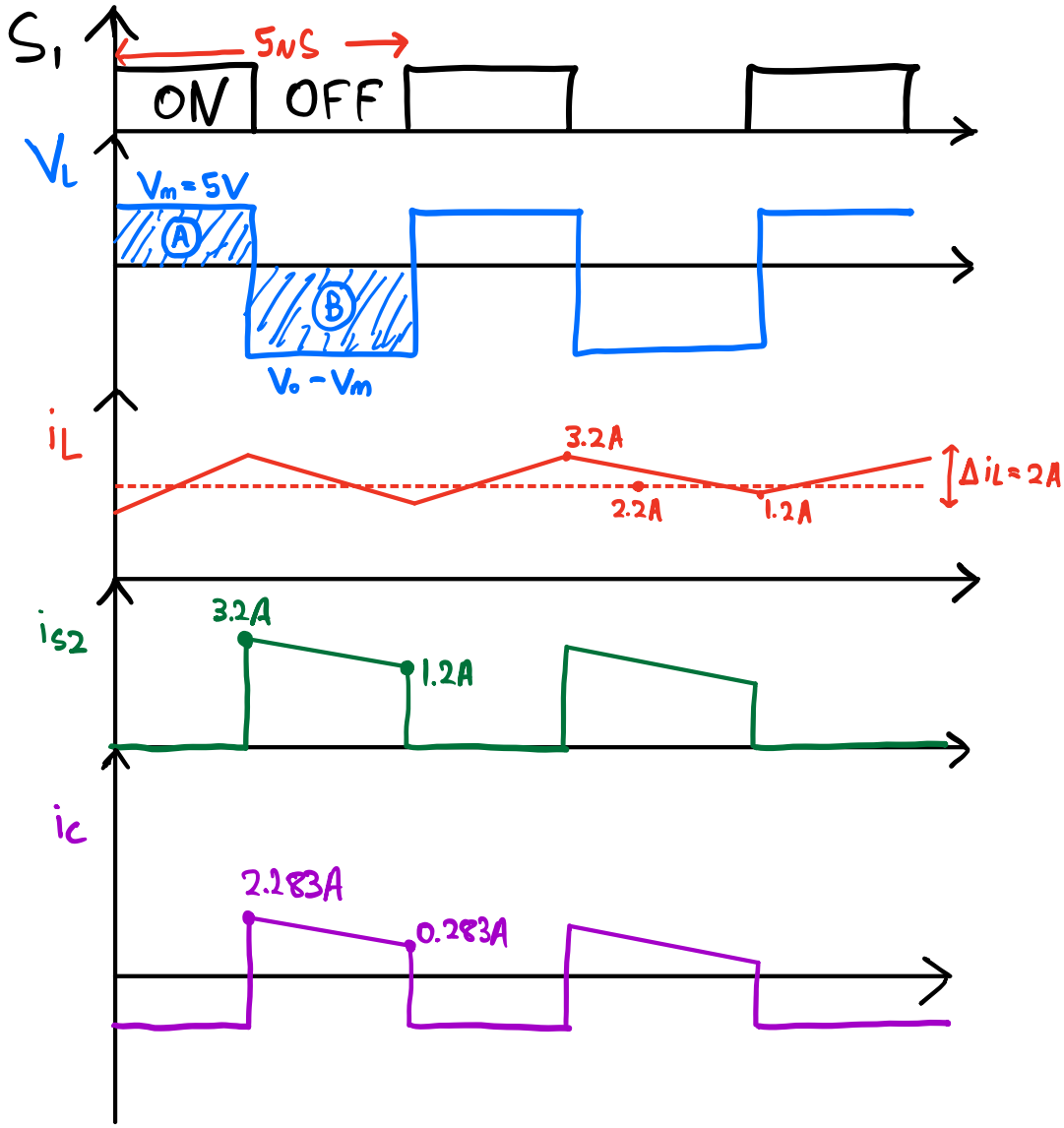
$$V_o = V_m \frac{1}{1-D} \quad \therefore D = 0.583$$

$$T_s = \frac{1}{200kHz} = 5\mu s$$

Area A = Area B

$$t_{on} \cdot V_m = t_{off} \cdot (V_o - V_m)$$

$$P_m = P_o = 11W$$

$$V_o I_m = V_o I_o$$

$$I_m = \frac{11W}{5V}$$